

Problem 1

$$c i) \quad L_{\pi_{\theta_1}}(\pi_{\theta_1}) = h(\pi_{\theta_1}) + \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \pi_{\theta_1}(a|s) A^{\pi_{\theta_1}}(s, a) - 0$$

$$A^{\pi_{\theta_1}}(s, a) = Q^{\pi_{\theta_1}}(s, a) - V^{\pi_{\theta_1}}(s)$$

$$\sum_a \pi_{\theta_1}(a|s) A^{\pi_{\theta_1}}(s, a) = \sum_a \pi_{\theta_1}(a|s) Q^{\pi_{\theta_1}}(s, a) - V^{\pi_{\theta_1}}(s) = V^{\pi_{\theta_1}}(s) - V^{\pi_{\theta_1}}(s) = 0$$

if $\eta \neq \phi$

$$L_{\pi_{\theta_1}}(\pi_{\theta_1}) = \eta(\pi_{\theta_1}) \quad \#$$

c ii)

$$\nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) = \sum_s d_{\mu}^{\pi_{\theta}}(s) \sum_a \nabla_{\theta} [\pi_{\theta}(a|s)] A^{\pi_{\theta}}(s, a)$$

$$\nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta})|_{\theta=\theta_1} = \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s)|_{\theta_1} A^{\pi_{\theta_1}}(s, a) - 0$$

$$\nabla_{\theta} \eta(\pi_{\theta}) = \sum_s d_{\mu}^{\pi_{\theta}}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s) Q^{\pi_{\theta}}(s, a)$$

$$Q^{\pi_{\theta_1}}(s, a) = A^{\pi_{\theta_1}}(s, a) + V^{\pi_{\theta_1}}(s)$$

$$\Rightarrow \nabla_{\theta} \eta(\pi_{\theta})|_{\theta_1} = \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s)|_{\theta_1} [A^{\pi_{\theta_1}}(s, a) + V^{\pi_{\theta_1}}(s)]$$

$$= \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s)|_{\theta_1} [A^{\pi_{\theta_1}}(s, a) + \sum_s d_{\mu}^{\pi_{\theta_1}}(s) V^{\pi_{\theta_1}}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s)]$$

$$\sum_a \pi_a(a|s) = 1$$

$$\Rightarrow \nabla_{\theta} \eta(\pi_{\theta})|_{\theta_1} = \sum_s d_{\eta}^{\pi_{\theta_1}}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s)|_{\theta_1} A^{\pi_{\theta_1}}(s, a) - \theta$$

① = ②, proved.

Problem 2

(a)

$$\min_{\theta} -g^T(\theta - \theta_k) \quad \text{s.t.} \quad \frac{1}{2}(\theta - \theta_k)^T H(\theta - \theta_k) \leq \delta.$$

$$\mathcal{L}(\theta, \lambda) = -g^T(\theta - \theta_k) + \lambda \left[\frac{1}{2}(\theta - \theta_k)^T H(\theta - \theta_k) - \delta \right]$$

$$\nabla_{\theta} \mathcal{L} = -g + \lambda H(\theta - \theta_k) = 0 \Rightarrow \theta(\lambda) - \theta_k = \frac{1}{\lambda} H^{-1} g$$

$$D(\lambda) = \min_{\theta} \mathcal{L}(\theta, \lambda) = \mathcal{L}(\theta(\lambda), \lambda)$$

$$= -g^T \frac{1}{\lambda} H^{-1} g + \lambda \left[\frac{1}{2\lambda^2} g^T H^{-1} g - \delta \right]$$

$$= -\frac{1}{\lambda} g^T H^{-1} g + \frac{1}{2\lambda} g^T H^{-1} g - \lambda \delta = -\frac{1}{2\lambda} g^T H^{-1} g - \lambda \delta$$

$$-\frac{g^T H^{-1} g}{2\lambda^2} + \delta = 0 \Rightarrow \lambda^* = \sqrt{\frac{g^T H^{-1} g}{2\delta}}$$

cb)

$$\theta^* - \theta_k = \frac{1}{\lambda^*} H^{-1} g \Rightarrow \theta^* = \theta_k + \frac{1}{\lambda^*} H^{-1} g$$

$$\Rightarrow \theta^* = \theta_k + \alpha H^{-1} \nabla L_{\theta_k}(\theta) \big|_{\theta_k} \quad \text{其中 } \alpha = \frac{1}{\lambda^*} = \sqrt{\frac{2\delta}{g^T H^{-1} g}}$$

Problem 3

L^{clip} 只在 A_t 相反的方向超出閾值時 (case 3, 4) 才停止梯度, 其它方向如 case 1, 6 允許有利於目標的更新無限制地繼續推進

L^{clip} 無論是有利還是不利, 只要對 γ 超出 $[1-\epsilon, 1+\epsilon]$ 就被硬性截斷, 因此也會阻止一些對優化實際有益的更新

1. Introduction

We implemented three algorithms:

1. **Vanilla DDPG** on *Pendulum-v1*.
2. **Vanilla DDPG** on *HalfCheetah-v4* (`ddpg_cheetah.py`).
3. **DDPG+Clipped Double Q (CDQ)** per TD3 on *HalfCheetah-v4* (`ddpg_cdq_cheetah.py`).

2. Architecture & Hyperparameters

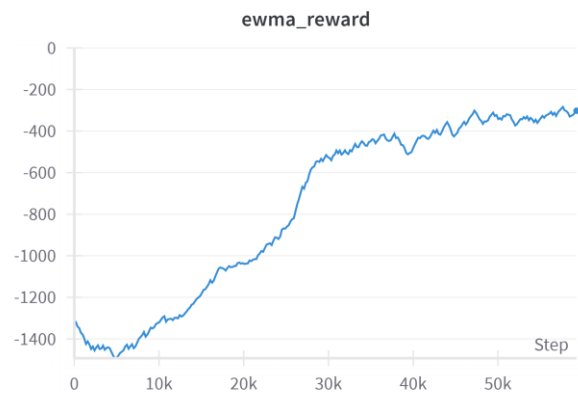
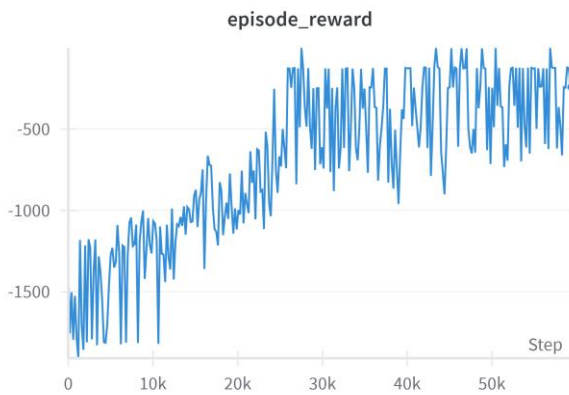
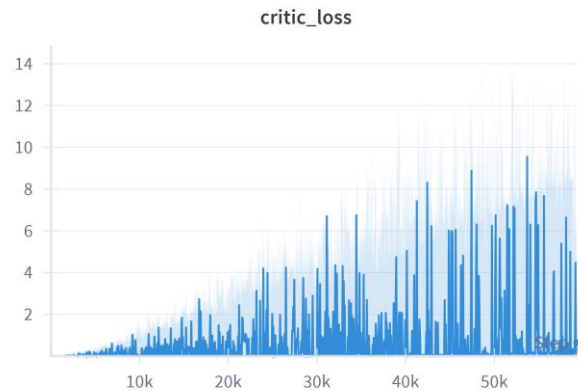
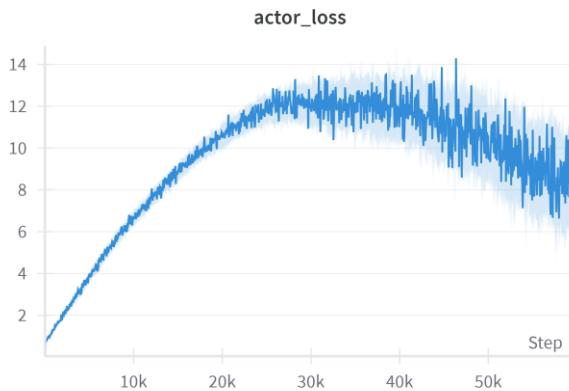
- **Actor & Critic:**
 - Two hidden layers (128–512 units), ReLU activations; tanh output for actions.
 - LayerNorm added in CDQ actor for smoother gradients.
- **Replay Buffer:** capacity = 1×10^6 , batch size = 256.
- **Exploration Noise:** Ornstein–Uhlenbeck ($\sigma=0.2$, $\theta=0.15$), clipped to `[action_low, action_high]`.
- **Optimizers:** Adam
 - Pendulum: `lr_actor=1e-4`, `lr_critic=5e-4`; $\tau=0.001$; $\gamma=0.99$.
 - HalfCheetah DDPG: `lr_actor=2e-4`, `lr_critic=1e-4`; $\tau=0.001$; $\gamma=0.995$.
 - CDQ: `lr_actor=1e-4`, `lr_critic=1e-3`; $\tau=0.005$; $\gamma=0.99$.
- **Gradient Clipping:** max norm = 1.0 for stability.

3. Experimental Protocol

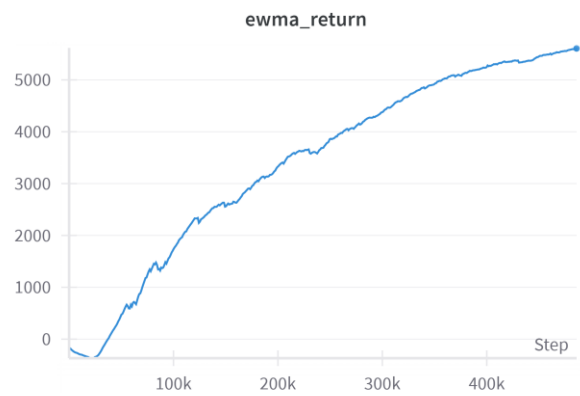
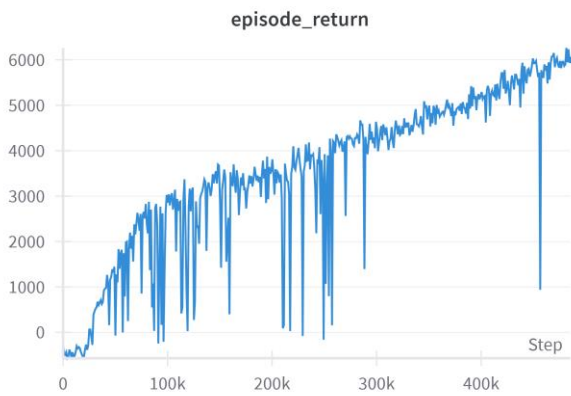
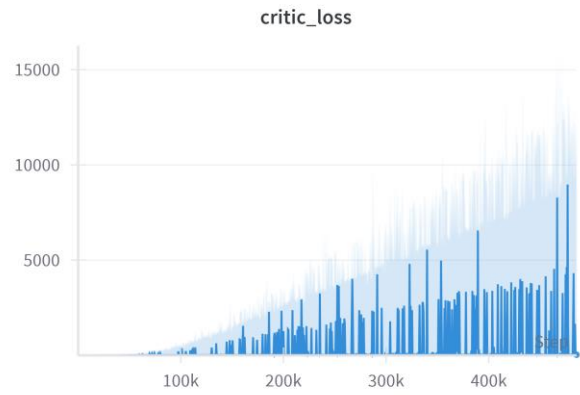
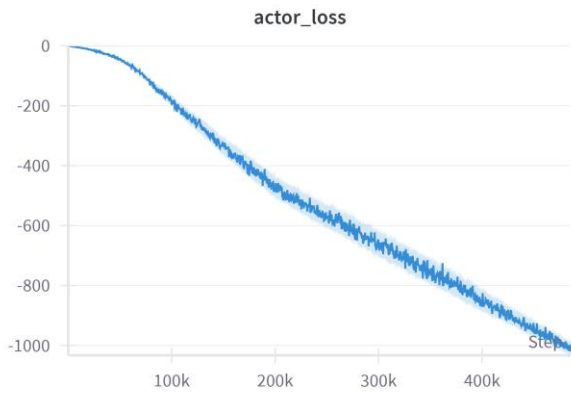
- **Pendulum-v1:** 300 episodes; warm-up = 1 024 steps; update each step thereafter.
- **HalfCheetah-v4:** 500 k environment steps; random policy for first 15 k (DDPG) or 25 k (CDQ) steps; then learn online.
- **Logging:**
 - **Losses:** critic & actor per update.
 - **Returns:** episode return and EWMA reward every episode (Pendulum) or every 10 k steps (HalfCheetah).

4. Results

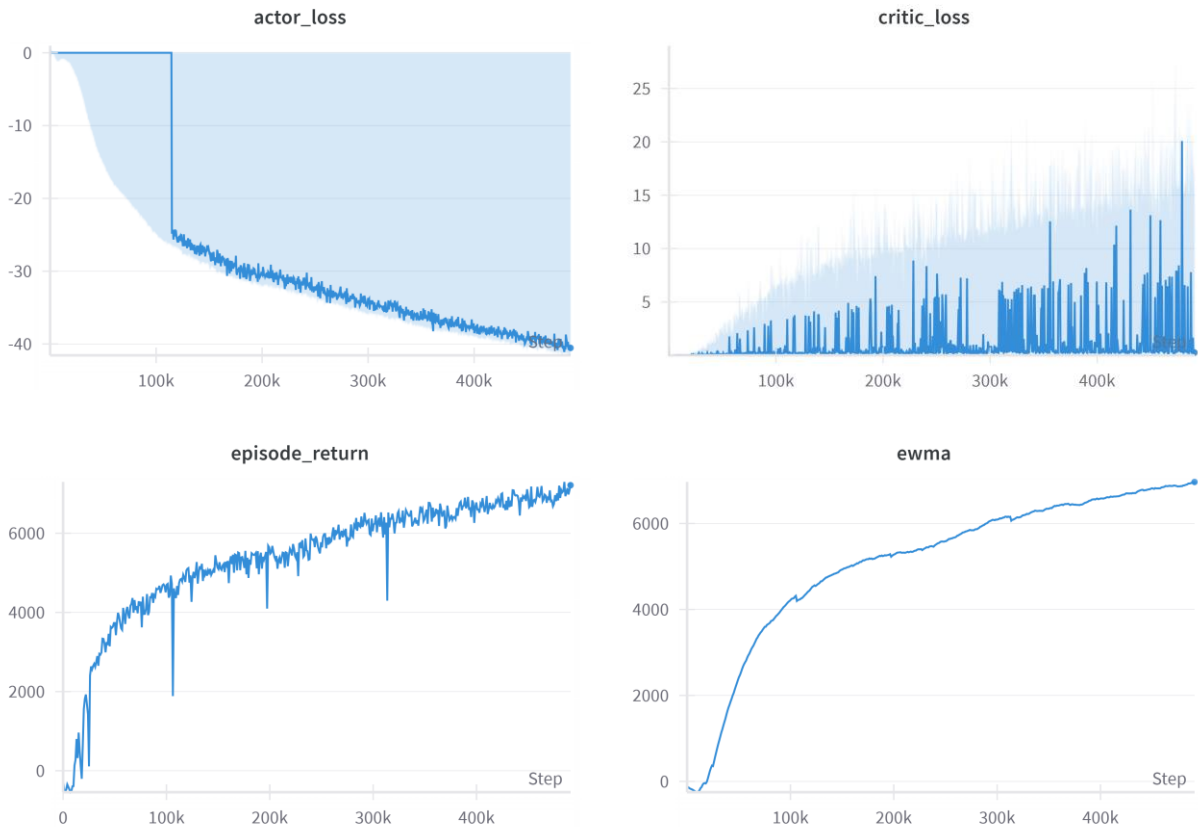
Task	Final Performance	Notes
Pendulum-v1	EWMA ≈ -200 by 200 ep	Rapid convergence; low variance.
DDPG Cheetah	Avg. return ≈ 6000	Moderate variance in returns.
CDQ Cheetah	Avg. return ≈ 7200	More stable learning; less overest.



(Pendulum-v1)



(DDPG Cheetah)



(CDQ Cheetah)

5. Discussion

- **Pendulum**: Vanilla DDPG suffices—policy settles quickly with minimal tuning.
- **HalfCheetah**:
 - Vanilla DDPG achieves baseline performance but suffers from Q-value overestimation and high variance.
 - **Clipped Double Q** (target smoothing + dual critics) reduces overestimation bias, stabilizes critic loss, and yields $\approx 30\%$ higher final returns.

6. Conclusion

1. **DDPG** can solve low-dimensional tasks with light hyperparameter tuning.

2. **Clipped Double Q** significantly boosts performance and stability in complex, high-dimensional control.
3. Further work:
 - a. Ablation on noise parameters ($\epsilon_{\max}$, σ) and τ schedules.
 - b. Comparison with full TD3 and SAC baselines.